

UTrucking AI Phone Assistant — QA & Testing Log

Prepared: 2026-07-02 **Agent:** Utrucking Agent (Retell AI) · versions v29 → v34 **How tested:** scripted conversations run against the *live* agent via Retell's playground API, one real phone call, direct probes of the lookup backend + Google Sheets, and edge-case audits of the new business endpoints (`/quote` , `/availability` , `/dispatch_plan` , `/billing_audit` , `/photo_quote`) and the customer estimate page.

1. Summary

The assistant was tested across **three layers**: (1) does it behave correctly turn-by-turn, (2) does it hold up to hard/ambiguous names at scale, and (3) does it work on a real call. Headline results:

- **10 / 10 functional scenarios passed** (order lookup, privacy gate, disambiguation, FAQ, transfer, call-end, identity verification).
 - **Hard-name stress test:** a garbled name **never** resolved to the wrong student (0 wrong matches across the sample).
 - **1 live phone call** completed successfully end-to-end; one minor wording tic was found and fixed.
 - **1 privacy risk** in the backend (over-eager name matching) was identified **and mitigated** by an added identity-verification step.
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2. Functional / behavior tests

Each scenario was driven as a real conversation against the published agent.

#	Scenario	Expected	Result
1	Caller gives name, checks order	Confirms identity, then answers	✓ Pass
2	One-question-at-a-time answers	Answers only the field asked, briefly	✓ Pass

#	Scenario	Expected	Result
3	Order mentioned <i>before</i> a name is given	Asks for the name first, no premature lookup	✅ Pass (fixed in v30)
4	Privacy gate — wrong/close name match	Confirms the name; reveals nothing if "that's not me"	✅ Pass
5	Ambiguous name → multiple matches	Offers choices, lets caller pick	✅ Pass
6	General question (pricing/services)	Brief, accurate answer from knowledge base	✅ Pass
7	Caller says goodbye	Warm close, then ends the call	✅ Pass
8	Caller asks for a person	Connects to the UTrucking team (transfer)	✅ Pass
9	Name not found after spelling	Escalates to the team instead of looping	✅ Pass (fixed in v32)
10	Identity verification	Confirms a second detail; if wrong, does not share and transfers	✅ Pass (added in v33)

3. Live phone-call test

A real call was placed to the assistant (caller: a real student record). The assistant fuzzy-matched a mispronounced name, confirmed identity, and answered pickup location, status, order ID, billing, delivery, and website questions — each concisely. **Finding:** the assistant occasionally tacked on filler ("right?", "is that okay?"). **Action:** fixed in v32. Full transcript: `utrucking-test-calls.txt`.

4. Name-matching stress test

Automated audit against the live roster (~1,655 students).

A. Misspelled hard-to-pronounce names (18 tested) — a letter-swap was applied to real names to simulate speech-to-text errors: - **10** matched to the correct student exactly - **8** returned "let me confirm which one" (the assistant then disambiguates) - **0** matched the **wrong** student - **0** failed to find anything

B. Fake names not in the system (12 tested) — should never match a real student: - **10** correctly rejected - **2** were over-matched to a real student by the backend (~17%) - **Mitigation:** the v33 identity-verification step blocks these — the caller cannot confirm a stranger's building, so no data is shared. **Root fix:** tighten the backend match threshold (planned).

5. Integration / infrastructure checks

Check	Result
Backend reachable (<code>/lookup_student</code> , <code>/health</code> , <code>/debug_sheets</code>)	✅ Online
Dispatch Google Sheet	✅ ~1,655 rows, all expected columns present
Service Google Sheet	✅ Fixed — 654 rows load via the gviz endpoint (was a wrong CSV URL, <i>not</i> an empty sheet)
Retell tools wired (<code>lookup_student</code> , <code>get_quote</code> , <code>check_availability</code> , <code>transfer_to_office</code> , <code>end_call</code>)	✅ Verified (v34)
Guardrails (jailbreak/abuse protection)	✅ Enabled (v33)

6. Known issues & mitigations

1. **Backend over-matching** (fake name → real student): mitigated by identity verification **and** permanently fixed — the match cutoff was raised to **0.6** in the backend.
 2. **Service sheet "empty":** ✅ resolved — the item/invoice sheet now loads **654 rows**; the "0 rows" was a wrong CSV URL, corrected to the gviz endpoint.
 3. **No phone number provisioned yet:** tested via the Retell dashboard/API; a live line can be added when ready.
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7. Business-engine & customer-tool audit (2026-07-02)

The Wave A/B/C endpoints and the new customer estimate page were audited with normal inputs, edge cases, and deliberately bad input. All were probed live.

Endpoint / tool	Test	Result
/quote	"five boxes and a mini fridge"	✓ \$133, itemized
/quote	structured list with an unknown item	✓ prices known items, lists the unknown as unmatched
/quote	empty / gibberish text	✓ returns \$0 gracefully, no crash
/availability	peak day (May 7)	✓ reports full , steers to nearest open day
/availability	capacity override	✓ respects the passed capacity
/availability	unreadable date	✓ now returns a friendly re-ask (fixed)
/dispatch_plan	peak day	✓ 126 stops · 36 buildings · 6 crews, clustered
/billing_audit	full sheet	✓ 24 flagged (15 missing order-id, 8 \$0/missing total, 2 missing invoice)
/photo_quote	image → items → price	✓ vision key authenticates; detects items and prices them
/estimate (customer page)	photo upload or typed items	✓ built — returns an itemized estimate + total
/lookup_student	fake name	✓ not_found (no false match — identity gate holds)

Bugs found and fixed this pass

1. **Quote parser dropped items with larger number-words.** *"twenty boxes and three mini fridges"* priced only the fridges (\$69) and **silently dropped the 20 boxes** — the worst failure mode on a customer estimate. **Fixed:** the parser now understands number-words to 99 and "a dozen", and a bare item defaults to **qty 1**, so nothing is ever dropped. Re-tested: the same input now returns **\$509**.
2. **Photo-quote leaked the API key in errors.** On a vision error the public `/photo_quote` endpoint echoed the full AI key in the message. **Fixed:** the key now travels in a request header (never a URL) and is redacted from any error text.

3. **Unreadable dates were silent.** `/availability` returned a blank record for an unparseable date. **Fixed:** it now returns a clear "what day were you thinking?" prompt.

Photo-estimate — live end-to-end proof

A random moving-boxes photo (a public image the system had never seen) was run through the full customer pipeline — **fetch image** → **AI vision (`gemini-2.5-flash`)** → **item detection** → **catalog pricing**:

- **AI detected:** 9× box, a dolly, 2× moving blanket, packing tape, a moving strap.
- **Quote returned:** 9× **UTrucking Box = \$198.00**, with the dolly / moving blanket / packing tape / moving strap correctly listed as *not priced* (they aren't stored items).

This confirms the customer *photo* → *instant estimate* flow works on unseen, real-world photos — not just curated test cases.

Photo-path hardening (surfaced by that test)

Fix	Why it matters
Switched vision model to <code>gemini-2.5-flash</code>	<code>gemini-2.0-flash</code> 's free-tier quota was returning <code>429</code>
Browser User-Agent + HTTP-status check on <code>image_url</code> fetch	Some hosts (e.g. Wikimedia) <code>403</code> a request with no User-Agent; the backend was sending an error page to the vision model
Real image mime-type sniffing (magic bytes)	Was hard-coded to JPEG — an iPhone HEIC or PNG upload would have failed
Browser-side downscale + JPEG conversion on upload	Faster uploads; normalizes HEIC and oversized photos
Generic AI names mapped to the catalog	"cardboard box" / "refrigerator" / "storage bin" now resolve to your items
Auto-retry on transient <code>429</code> / <code>503</code>	The vision API briefly overloaded mid-test; a retry recovered it
API key moved to a request header + redacted from errors	The public endpoint had been echoing the key on an error

8. Method note

Behavior was validated by replaying full conversations against the **live agent** (not a mock) and inspecting every assistant message and tool call. Name matching and the business endpoints were audited directly against the production backend and Google Sheets. Testing spanned agent versions v29 through v34; each fix was re-tested before publishing.